

CMOS FABRICATION

LECTURE – 4 :: (01-04-2025)

Er. Sailesh Mishra

[Spec. Microelectronics & VLSI]

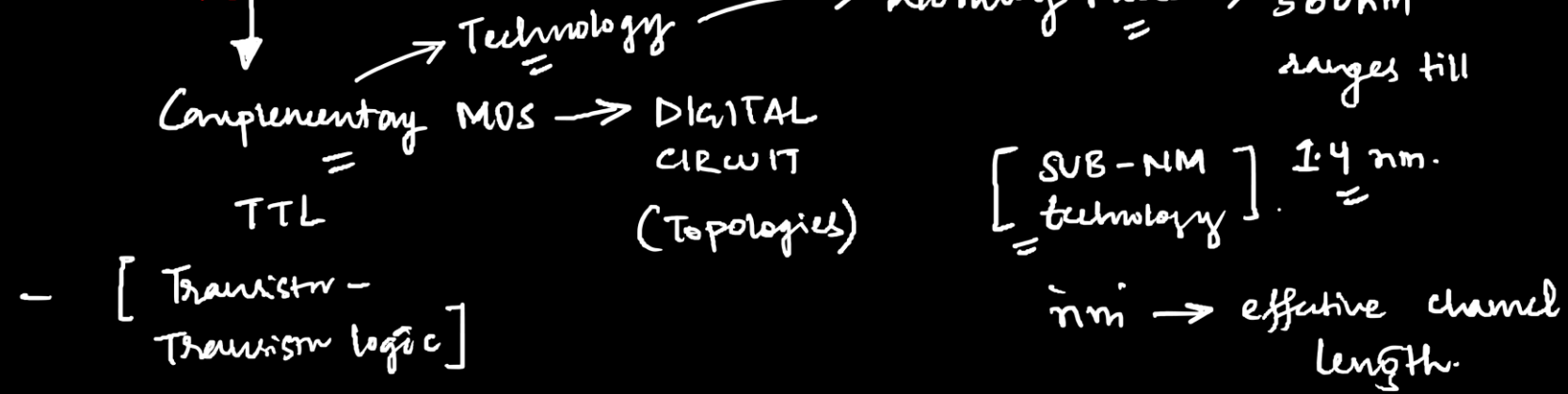
Silicon University, Bhubaneswar

Q. WHY WE'RE FABRICATING CMOS?

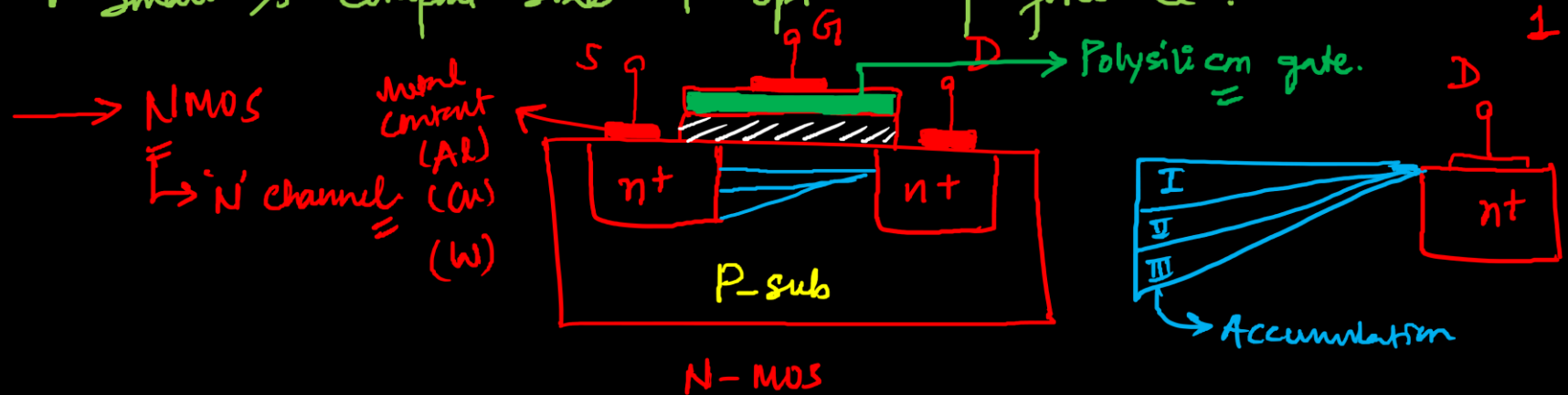
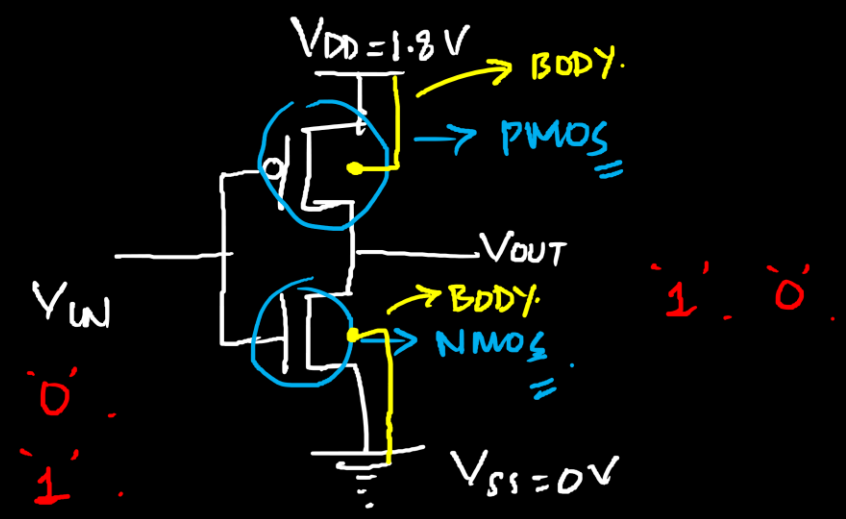
Because, NOT gate is an UNIVERSAL gate !!

- It has Better Switching Speed
- good i/p impedance; stronger NOISE MARGIN / NOISE Tolerance.
- low power Consumption.
- Easier Fabrication & Testing
- Highly controllable & Versatile
- Technology viable and wide Bandwidth of Applications.
- Small & Compact size. + optimum performance.

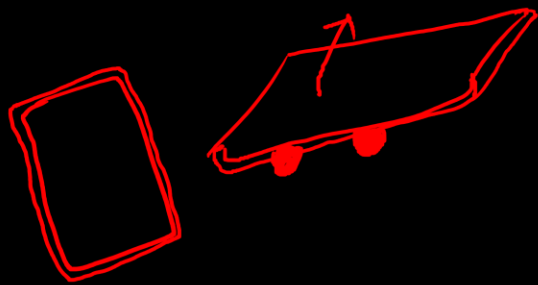
CMOS FABRICATION



ECL
Emitter coupled logic



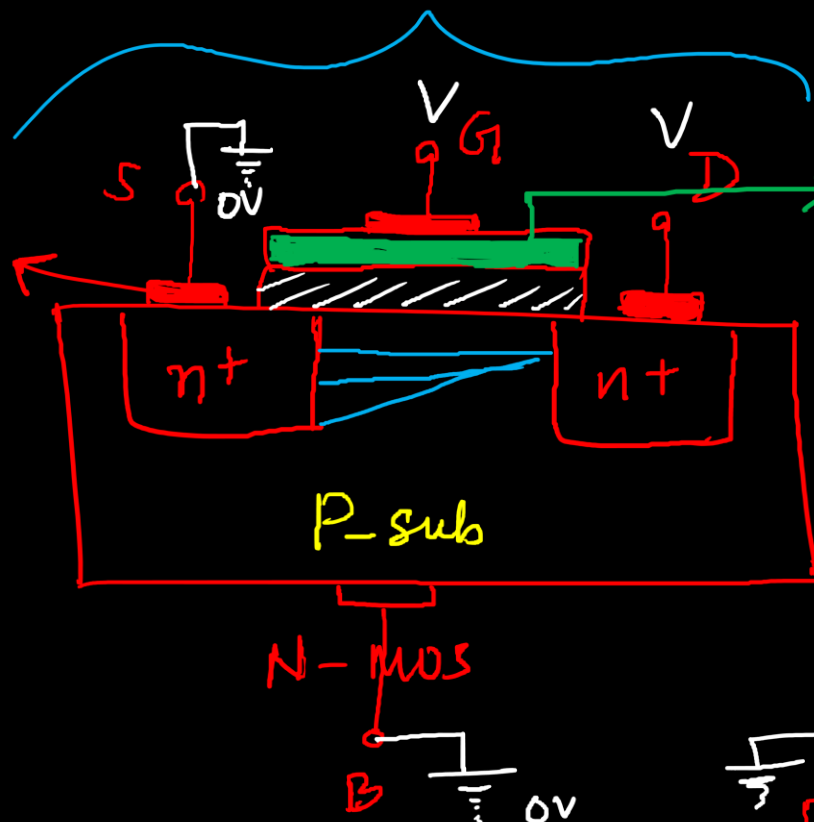
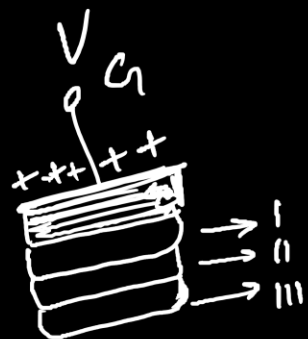
2nd yr.
NOT gate (INVERTER).



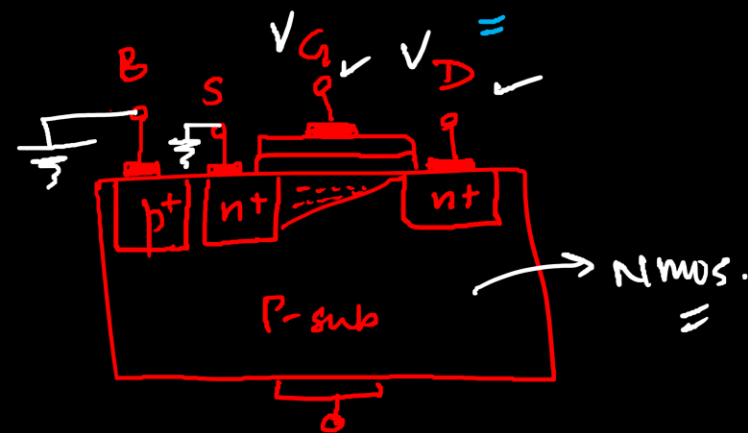
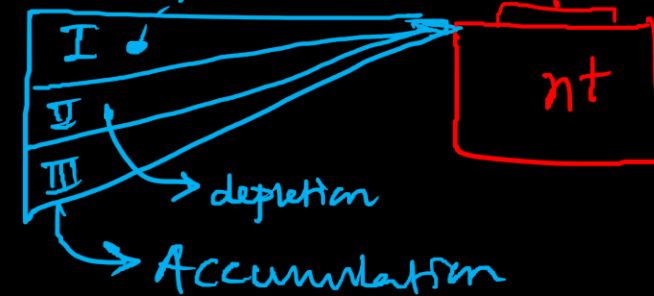
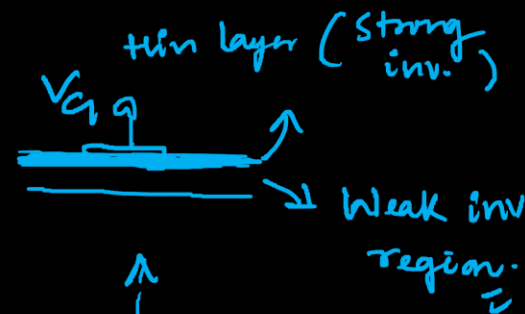
S, G, D
3-terminals.

→ NIMOS

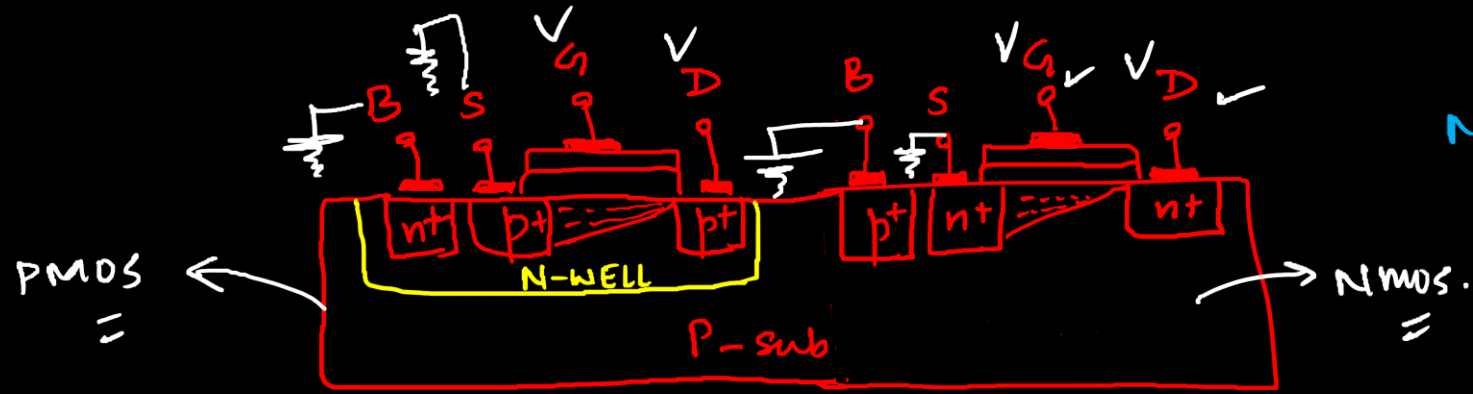
→ 'N' channel
= (AR)
= (W)



Polysilicon gate.

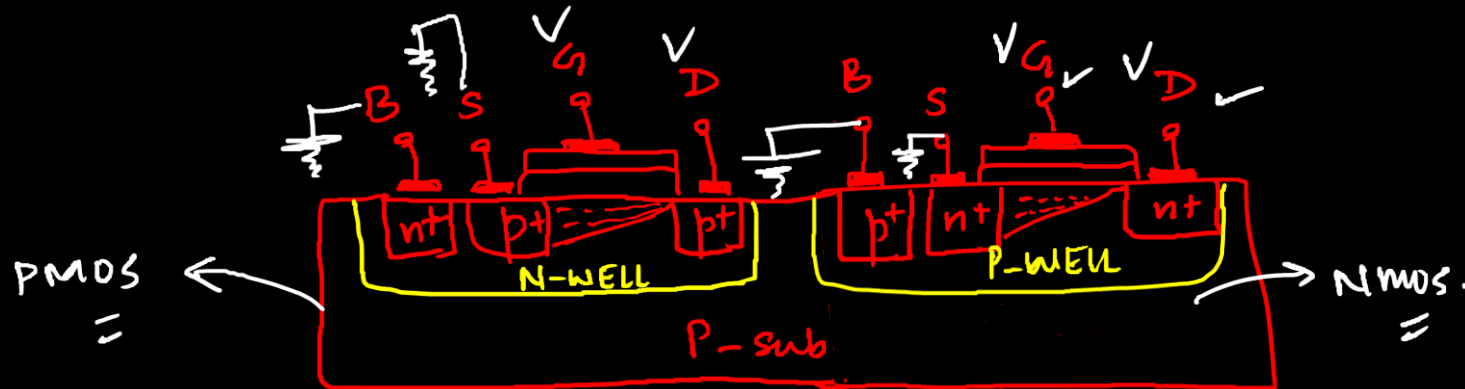


①.



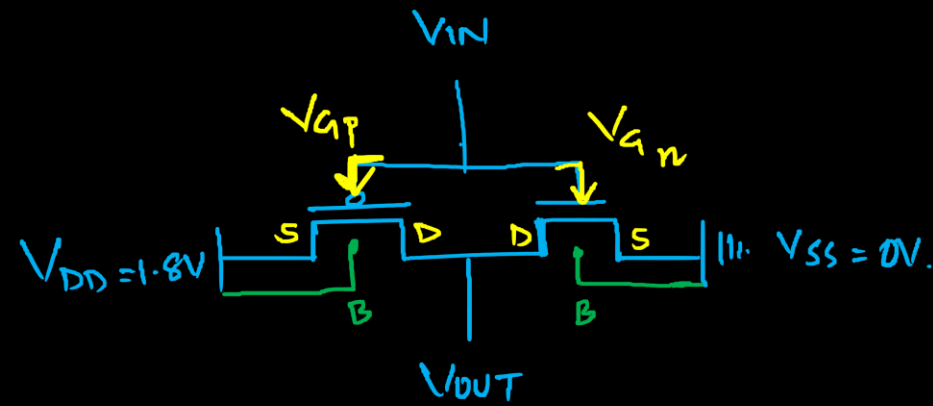
N-WELL CONCEPT
OF FABRICATION

②.

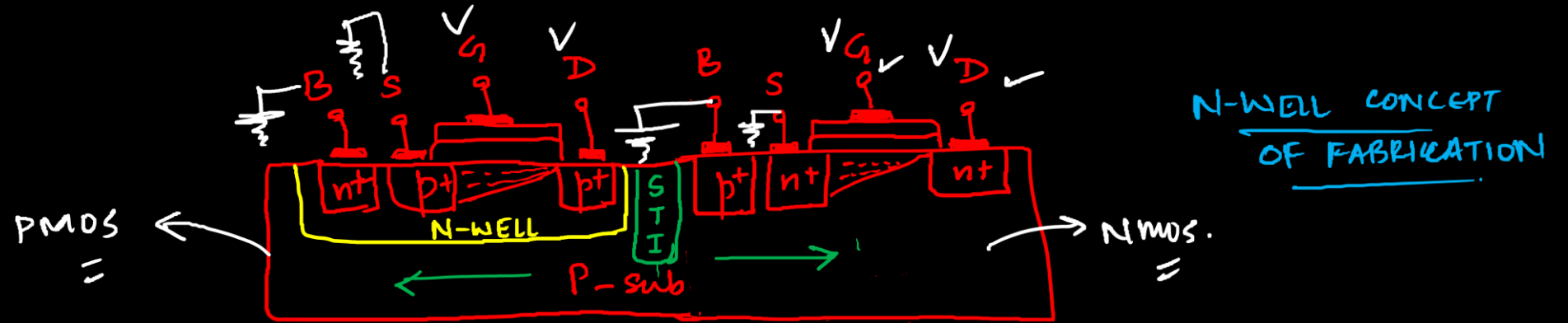


TWIN-WELL CONCEPT
OF FABRICATION

CMOS
VLSI
DESIGN

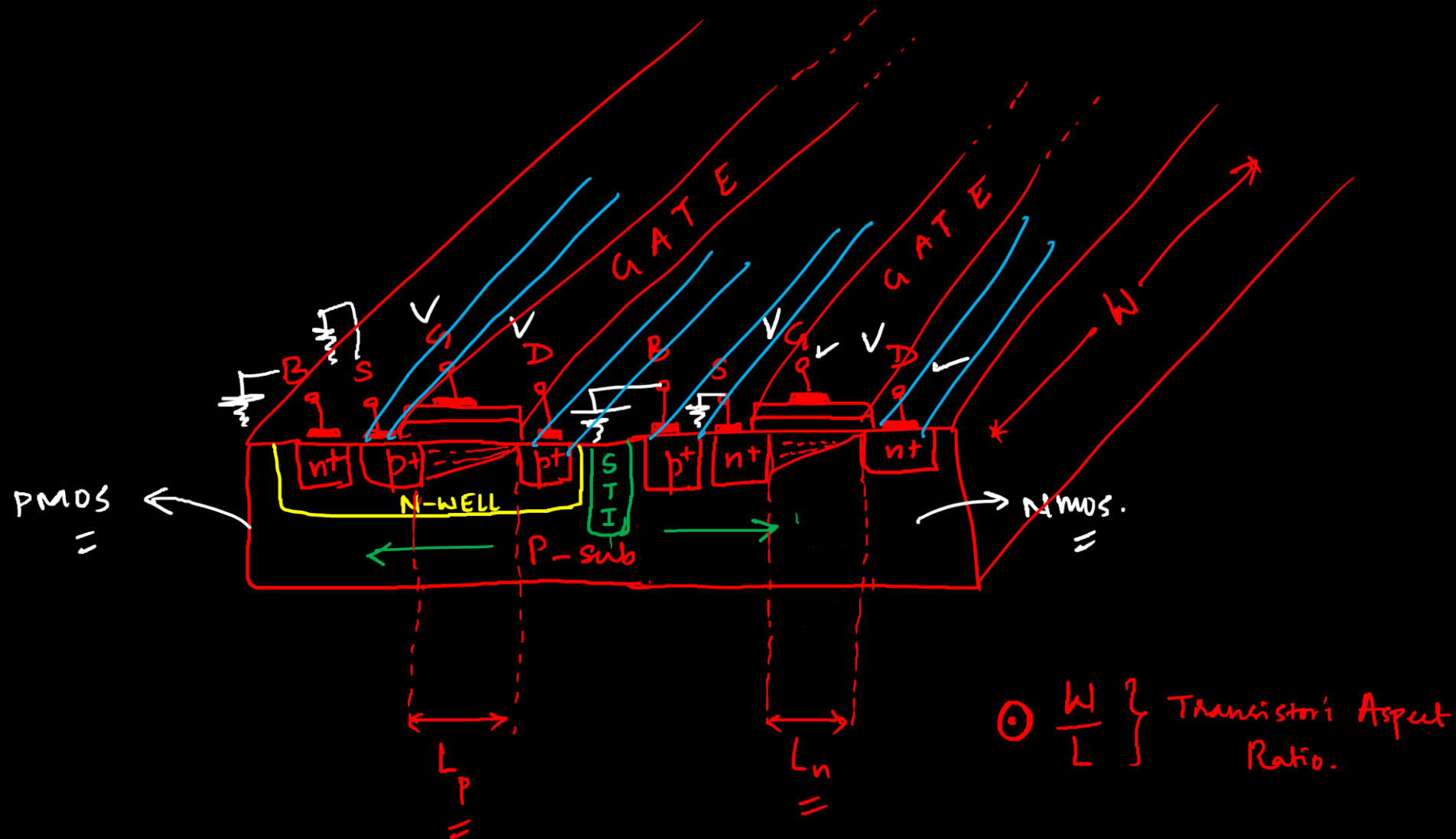


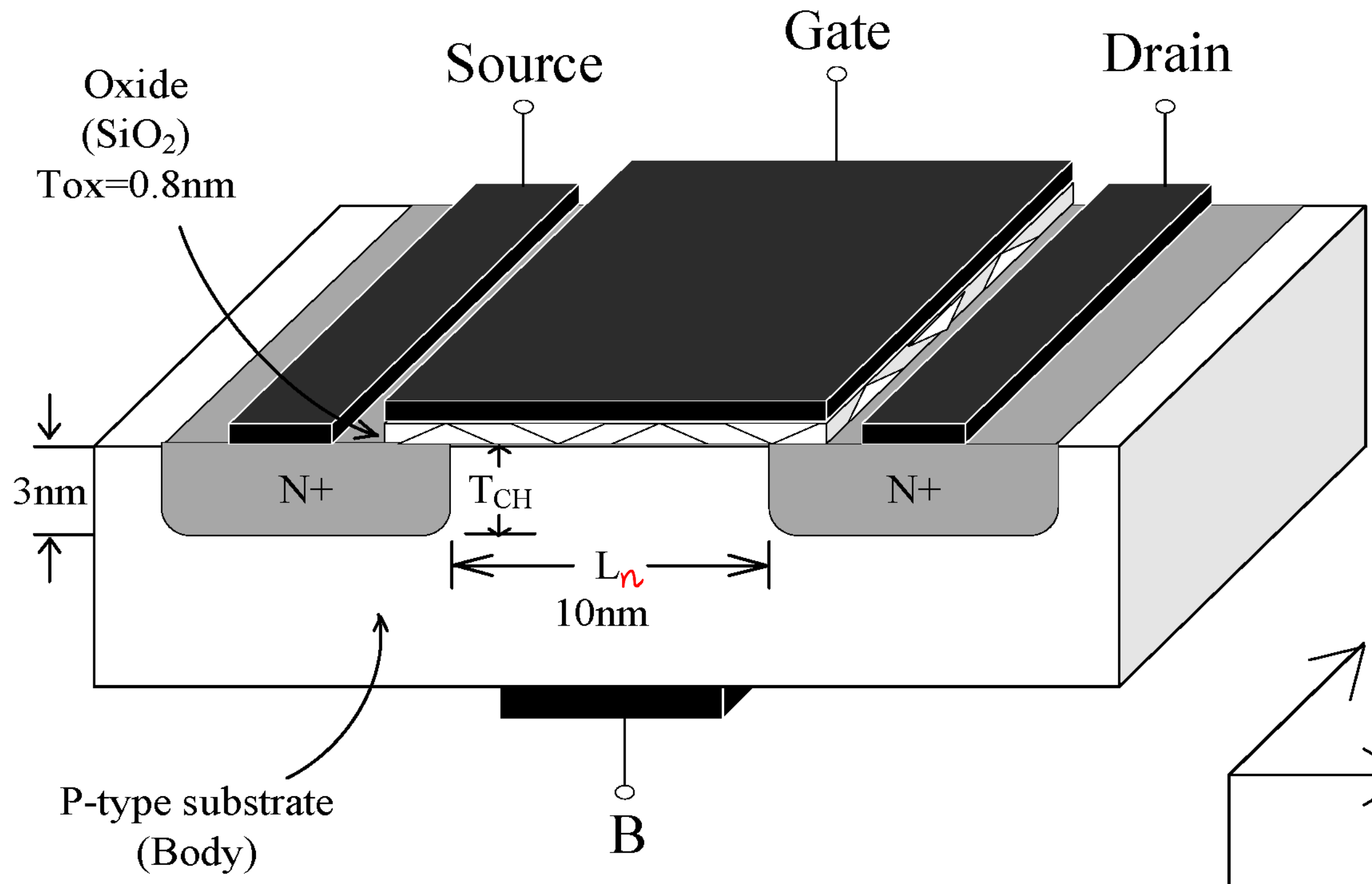
①.



SHALLOW TRENCH ISOLATION $\rightarrow$ STI $\rightarrow$ provides isolation to the CMOS from forming UNWANTED CHANNEL.

[DTI] $\times$ $\rightarrow$ Complex Fabrication process.





NMOS

PMOS

